CLOUD DEPLOYMENT MODELS

Use Case, Comparison and Recommendation

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Subject: Cloud Computing
Chosen use case: Zomato-like food ordering app

This assignment describes the technical requirements of a food delivery app, compares public, private and hybrid cloud models, recommends a suitable model, and considers a sudden increase in users.

Task 1: Use Case and Technical Requirements

Use case: A Zomato-like food ordering app lets customers browse restaurants, view menus, place orders, pay online, and track delivery. Restaurants update menu availability and delivery partners receive order details.

Main technical requirements:

• **Compute and application hosting:** servers or containers to run the mobile app backend, restaurant portal and APIs.

• **Database and storage:** a database for users, restaurants, orders and payment status; object storage for restaurant and food images.

• **Network and delivery:** secure internet connectivity, load balancing, HTTPS/TLS and a content delivery network (CDN) for images and static files.

• **Security:** identity and access management, encryption, secure payment-provider integration, protection of customer addresses and contact details, and regular patching.

• **Availability and scaling:** monitoring, backups, recovery plans and automatic scaling for lunch/dinner rushes, promotions and festival periods.

• **Maintenance:** logging, alerts, deployment pipelines and testing so updates can be released with minimal downtime.

Task 2: Public vs Private vs Hybrid Cloud

Factor	Public Cloud	Private Cloud	Hybrid Cloud
Cost	Usually low upfront cost; pay for resources used, but poor monitoring can lead to unexpected bills.	High setup and maintenance costs; organization pays for dedicated capacity.	Costs of both environments and integration; can optimize by placing each workload appropriately.
Security	Strong cloud security tools are available, but the app team must configure access, encryption and network rules correctly.	More direct control over infrastructure and policies, but the organization must operate and secure it.	Sensitive workloads can stay in a controlled environment while other workloads use public cloud; integration must be secured.
Scalability	Can scale quickly for spikes such as dinner rush or discount campaigns.	Scaling may require purchasing and installing additional hardware.	Public cloud can absorb some spikes, while private systems continue handling selected workloads.
Ease of maintenance	Provider maintains physical infrastructure and many managed services; customer still maintains app and configurations.	Requires in-house staff to maintain hardware, platform and updates.	More complex because teams must maintain and connect two environments.
User data and traffic	Data can be stored in managed databases with access controls; traffic can use load balancers and CDN.	Data stays in organization-controlled infrastructure; public-facing capacity may be harder to expand quickly.	Can separate sensitive data or systems from elastic public-facing services, but data transfer must be carefully designed.

Task 3: Recommended Deployment Model

I recommend the **Public Cloud** for a new Zomato-like food ordering app. First, public cloud can scale compute and network resources quickly when order volume rises during meal times, festivals or promotions. Second, it usually has lower upfront cost than building a private data centre, and managed databases, monitoring and deployment services reduce infrastructure maintenance. The team must still secure customer data, restrict access, encrypt sensitive information and monitor spending.

Task 4: Hybrid Cloud During Sudden Popularity

One advantage: A hybrid design can keep selected sensitive systems or data in a controlled private environment while using public-cloud capacity to handle a sudden increase in app traffic, images and APIs.

One disadvantage: Connecting the two environments adds complexity. Network links, identity integration or data synchronization can become bottlenecks during the spike, and managing both environments may cost more.

Task 5: AI-Generated Summary for Indian Apps

Public Cloud: Often a good fit for public-facing services that need fast deployment and elastic capacity. For example, a BookMyShow-like ticketing app could use it to handle ticket-sale traffic spikes, media delivery and application APIs.

Private Cloud: May suit workloads needing more direct infrastructure control, specialized governance or strict internal requirements. A Paytm-like payments business might choose private environments for selected sensitive workloads where policy or compliance requirements justify them; this does not claim that Paytm actually uses only a private cloud.

Hybrid Cloud: Can suit large services that need both controlled environments and flexible public-cloud capacity. An IRCTC-like booking platform might keep selected core or regulated systems in a controlled environment while using public cloud for web front ends, analytics or peak-demand support, if its architecture and compliance needs permit.

Surprising insight: An app being available to everyone does not mean it must run entirely on public cloud. A company can combine deployment models, and the best choice may differ between its website, database, analytics and identity systems.

Conclusion: For a new food-ordering app, public cloud is a practical starting point because it scales quickly and reduces upfront infrastructure work. Hybrid cloud can be useful when there is a clear reason to keep some systems under tighter control, but it introduces extra integration and operational complexity.